Day 15 4/11/26

66. Plus One

Input: [1,2,3], Output [1,2,4]

Approach: Right-to-Left with Carry

The addition starts from the last digit, just like normal addition.

1. Start from the last index.
2. If the digit is less than 9, simply increment it and return the array.
3. If the digit is 9, adding 1 marks it 0, so set it to 0 and continue to the previous digits.
4. If every digits was 9, create a new array with 1 at the begining

Time complexity : O(n)
Space Complexity: O(1)

Code

```
class Solution{
  public int[] plusOne(int[] digits){
    for(int i=digits.length-1; i>=0; i--){
      if(digits[i]<9){
        digits[i]++;
        return digits;
      }
      digits[i]=0;
    }
    int[] result=new int[digits.length+1];
    result[0]=1;
    return result;
  }
}
```

Description | Accepted | Editorial | Solutions | Submr

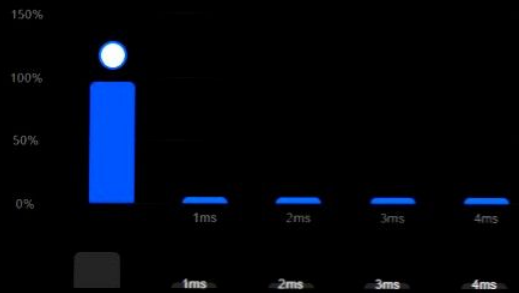
All Submissions

Runtime

0 ms | Beats 100.00%

Memory

42.96 MB | Beats 81.35%



Code Java

Code

Java | Auto

```java
class Solution {
    public int[] plusOne(int[] digits) {

        for(int i=digits.length-1;i>=0;i--){
            if(digits[i]<9){
              digits[i]++;
              return digits;
            }
            digits[i]=0;
        }
        int[] arr = new int[digits.length+1];
        arr[0]=1;
        return arr;
    }
```

Saved

Ln 13, Col 20

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2 | Case 3

Input

digits =

[1, 2, 3]